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EXAMEN CODE:

CUBA 2025
CONCURSO NACIONAL DE QUÍMICA

Specific instructions for the Chemistry Contest

Highly esteemed student:

- This exam was created by a Scientific Committee composed of former olympic students who work (mostly) at the University of Havana.
- **Complete your personal information**. Please verify with your ID card.
- The direction of the Provincial Education is responsible for ensuring that the exam is printed in a sufficiently high quality, allowing students to read the instructions, questions, figures, data, graphs, and tables. **No questions will be canceled due to poor printing.**
- Write all your information clearly in the space provided (first name, last name, school, municipality, province). **Do not write anything in the CODE field**, it is for the exclusive use of the coding manager.
- **Check that your exam is complete.** In the case of the exam on the second day, 10th grade should answer problems 1, 2, 3 and 4; 11th grade should answer problems 1, 2, 3 and 5; 12th grade should answer problems 1, 2, 5 and 6.
- All necessary constants, conversions, equations, and periodic table are included on the first page of the exam. This page will be removed during the coding process.
- At the beginning of each question, the grading is indicated, its value (out of a total of 100 points), as well as the value of each item (in marks).
- You can use a scientific calculator and a ruler.
- Please comply to the rules of approximate calculation. Attention! You have limited time and the exam has a large number of items. You do not have a space limit to answer, BUT please answer in an orderly manner, with appropriately sized and legible font, to facilitate the grading process.
- You must present all auxiliary calculations, justifying your reasoning and answers properly whenever necessary. **Unintelligible answers will not be graded.**
- Number the additional answer sheets at the bottom of each page and indicate the total number of pages, for example, “1 of 7,” “2 of 7,” and so on.
- If the exam instructor/proctor interferes with any of the above guidelines due to ignorance, exercise your right to protest and immediately request the intervention of the competent authority of the Ministry of Education, which is the institution responsible for administering the exam.
- The President of the Scientific Committee can be reached at +53-5045-4299.

You are a source of pride for Cuban education: Best of luck!

EXAM CODE (only for the use of the code manager)

Avogadro's Constant	$N_A = 6{,}0221 \cdot 10^{23}\ \mathrm{mol^{-1}}$	Ideal Gas Law	$p \cdot V = n \cdot R \cdot T$
Gas constant	$R = 8{,}3145\ \mathrm{J \cdot K^{-1} \cdot mol^{-1}}$ $= 0{,}08205\ \mathrm{atm \cdot L \cdot K^{-1} \cdot mol^{-1}}$	First Principle of Thermodynamics	$Q = \Delta U + W$
Zero in Celsius scale	$273{,}15\ \mathrm{K}$	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485\ \mathrm{C \cdot mol^{-1}}$	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1{,}00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln\left(\frac{c_{red}}{c_{ox}}\right)$
Arrhenius' equation	$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (\nu - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{crio} = i \cdot K_{crio} \cdot b$
Absorbance	$A_\lambda = -\log_{10}\left(\frac{I}{I_0}\right) = \varepsilon \cdot l \cdot c$	Vapor pression lowering	$\Delta p = \frac{i \cdot x_s}{x_{dis} + \sum i \cdot x_s} \cdot p^\circ_{dis}$
Sphere volume	$V_{sp} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1\ \mathrm{bar}$ $c^\circ = 1\ \mathrm{mol \cdot L^{-1}}$
Sphere area	$A_{sp} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Circle area	$A_{sp} = \pi \cdot r^2$	Speed of light	$c = 2{,}998 \cdot 10^8\ \mathrm{m \cdot s^{-1}}$
Length of a circumference	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6{,}626 \cdot 10^{-34}\ \mathrm{J \cdot s}$
Gravity force	$F_g = m \cdot g$	Gravity on Earth	$g = 9{,}81\ \mathrm{m \cdot s^{-2}}$
1 J = 1 C·V = 1 A·s·V = 1 kPa·L = 1 Pa·m^3 = 1 N·m = 1 W·s = 0,2389 cal 1 N = kg·m·s^{-2}			
1 atm = 760 Torr = 760 mm = 101,325 kPa 1 bar = 100 kPa 1eV = 96,485 kJ·mol^{-1}			
π = 3,1416 1 pm = 0,01 Å = 10^{-12} m 1 dm^3 = 1 L 1 mL = 1 cm^3 1 M ≡ 1 mol/L			

	1	2	3		4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																		2 He 4.003
2	3 Li 6.941	4 Be 9.012												5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305												13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956		22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906		40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	*	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207,2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	**	104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]

EXAM CODE (only for the use of the code manager)

Problem 7. Carbon and sulfur (15% of total; 10th, 11th and 12th grades)

7.1	7.2	7.3	7.4	7.5	7.6	Total	Total
10	8	4	8	2	28	60	15

Ernesto is a very curious student, and whenever he has the opportunity, he conducts experiments in the school laboratory with technician Ivis and mentor Agustín. After conducting a bibliographic search, the mentor discovered that when carbon and sulfur combine, different binary substances are obtained, and he asked the student to answer the following question.

7.1. What is the formula for the four sulfur compounds of carbon containing approximately 36, 27, and 16% carbon by mass? It is known that in one case, both the monomer and the dimer can exist, and none of the molecules is made up of more than five atoms.

7.2. Help Ernesto to represent the structures of the four molecules described above.

Ernesto liked the challenge and asked permission to try to obtain the substances at the school. But Ivis denied him, arguing that sulfur compounds are toxic and that the reactions involved were highly exothermic. Fortunately, Agustín has excellent relations with the University of Sancti Spíritus, and they planned to perform the reactions under more controlled conditions the following day.

Upon arriving at the university, Ivis found a **10,0 L** rigid reactor equipped with a pressure gauge and placed it under a fume hood. Ernesto weighed 60,0 g of a carbon-sulfur mixture on a technical balance and placed it inside the container. At that moment, Agustín sent them out for a snack, and when he was left alone in the laboratory, he secretly modified the experimental conditions. Upon returning, the student pressed a button, producing a spark and a small explosion inside the reactor. After cooling to room temperature, Ernesto was shocked and disappointed because the pressure inside the container was back to where it had been. Upon careful opening, he noticed a very unpleasant odor and the absence of solid residue. Then the mentor confessed: he had removed most of the solid mixture and filled the container with oxygen.

7.3. Can you help Ernesto to formulate the chemical reactions that happened?

“Agustín, you’re always so reckless!” Ivis said angrily and expelled him from the room. Ernesto weighed the portion of the carbon-sulfur mixture his mentor had hidden. He found a mass of only **57,3 g** and poured it completely out of the reactor. The technician made sure to evacuate all the air and replace it this time with the inert gas argon, until a pressure of 1,00 atm was reached. A spark was then ignited, and a strongly exothermic reaction occurred inside the vessel, resulting in the formation of substances **A** and **B** in a gaseous state. No solid residue was observed through the glass window of the apparatus. The system was then placed in an ice bath. After cooling to 100 °C, the pressure inside the vessel was 3,81 atm. Cooling continued, and upon reaching the temperature of 25 °C, a few drops appeared on the window due to the condensation of some of product **B**, and the pressure dropped to 1,96 atm. Upon reaching 0 °C and after carefully evacuating the gases (only Ar and product **A**), between 37 and 38 mL of a clear, volatile liquid with a very unpleasant odor were recovered (all of product **B**), which, when measured with a hydrometer, had a density of 1,29 g/mL.

7.4. Show that the amount of oxygen initially introduced into the reactor by the mentor was sufficient to react with the small portion of the mixture that was combusted.

Vapor pressure is the pressure exerted by the portion of a volatile substance in the gaseous phase, in direct contact –that is, in equilibrium– with the corresponding volatile liquid or solid. Vapor is said to be “saturated” when the gaseous phase contains the largest amount of the volatile substance thermodynamically possible.

7.5. The saturated vapor pressure of **B** at 25 °C is 48,1 kPa. Calculate the maximum amount of **B** in the gas phase that the reactor can hold at that temperature.

7.6. Help the student to calculate the identity of **A** and **B**.

Note: The experiments described in this problem are entirely fictitious and extremely dangerous. **Never** attempt to reproduce them.

EXAM CODE (only for the use of the code manager)

Problem 11. Iodometry or iodimetry (20% of total; only 11th and 12th grades)

11.1	11.2	11.3	11.4	11.5	11.6	11.7	11.8	11.9	11.10	11.11	11.12	11.13	11.14	11.15	11.16	Total	Total
2	4	2	2	6	2	7	2	8	4	8	4	9	4	9	7	80	20

Mentor Orestes (better known by his surname Landrove) knows that iodine methods are suitable for performing quantitative determinations of certain analytes. Although it is possible to perform the titration directly using iodine, he always recommends using an alternative titrant, typically sodium thiosulfate, which is a mild reducer capable of reacting with the iodine formed by an analyte. The main reaction that occurs is

$$2S_2O_3^{2-}(ac) + I_2(ac) \rightarrow S_4O_6^{2-}(ac) + 2I^-(ac)$$

11.1. One of the reasons why iodine is not preferred as a titrant is that it is very volatile. To counteract this difficulty, it is common to add excess iodide ions. Why?

11.2. Calculate the equilibrium constant of the main reaction of the titration at 25 °C.

11.3. The reaction above is based on the high affinity of the sulfur in thiosulfate for iodine. Explain this phenomenon according to the Hard and Soft Acids and Bases Theory.

11.4. The stability of thiosulfate solutions is very sensitive to the pH of the medium. For example, acidic impurities cause cloudiness and a pungent odor. Write the equation for the reaction that occurs.

During the training prior to the International Chemistry Olympiad, Landrove prepared several experiments based on diiodine titrations for Hector. First, they attempted to determine the vitamin C (ascorbic acid, $C_6H_8O_6$, abbreviated as AA hereafter) content in the juice of various fruits. Since ascorbic acid (AA) is not generated through any human metabolic pathway, consuming foods rich in vitamin C, such as citrus fruits, tomatoes, and potatoes, is necessary to prevent diseases like scurvy, which is characterized by collagen deficiency. Hector decided to quantify AA using a redox titration with a standard iodine solution, forming iodide and dehydroascorbic acid ($C_6H_6O_6$). Starch is used as a visual indicator, which forms a deep blue compound in the presence of traces of iodine.

11.5. What masses of iodine and potassium iodide does Hector need to weigh to prepare one liter of aqueous solution with molar concentrations of 0,00100 M and 0,0500 M, respectively?

11.6. Formulate the equation for the reaction between iodine and ascorbic acid.

11.7. In general, the pH of the I_2 solution is adjusted to avoid disproportionation of this species. Which medium is preferred? Justify numerically.

Landrove macerated a variety of fruits and vegetables separately in distilled water using a mortar and pestle. He then filtered the mixture, washed the solid portion, and transferred the collected liquid to a 50-mL flask, topping up with additional filtrated liquid to the required volume. Hector then transferred 10,0 mL of the solution to an Erlenmeyer flask using a volumetric pipette. He added approximately 20 mL of distilled water and 2 mL of 1% starch and titrated with the 0,00100 M iodine standard solution until the first sign of a permanent blue color appeared. The results are in the table.

Food	orange	plump	pepper	guava	parsley
$V(I_2)$, mL	12,5	9,8	16,2	20,0	8,6

11.8. What glassware is needed to perform a gravity filtration?

11.9. Typically, AA content ranges from 10-30 mg per 100 mL of orange juice. Do Hector's results correspond to the values reported in the literature?

Landrove then explained to Hector that the concentration of the standard solution he had prepared was probably incorrect and suggested using back titration to determine the ascorbic acid content. These methods generate an excess of iodine by reacting with an oxidizing primary standard. Some of the iodine formed reacts with the analyte, and the excess is titrated with thiosulfate.

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11.10. Formulate the reactions corresponding to the reactions of iodide ions with dichromate and iodate ions in an acidic medium.

11.11. The guava juice that Hector analyzed contained 35,2 mg of AA per 100 mL. Considering that only AA contributes to the acidity of the food, estimate the pH of the solution.

At the end of the session, the mentor tested the student's knowledge of chemical equilibrium. Landrove explained that the copper(II) ion is not capable of oxidizing iodide ions, as evidenced by the standard electrode potential values. However, the mentor dissolved some copper sulfate pentahydrate crystals in pure water in a test tube. Upon adding a portion of potassium iodide and stirring, the solution immediately changed color, first to green and then to dark yellow, and a precipitate appeared. After filtering and washing with water, the precipitate turned white.

11.12. Help Hector explain why the redox reaction of copper(I) iodide precipitate formation occurs, taking into account the polarization of ions and the Hard and Soft Acids and Bases Theory.

11.13. Hector found it easy to demonstrate numerically that in the presence of excess iodide ions, the process described above is enhanced. Can you do it?

11.14. Calculate the solubility (in mol/L) of copper(I) iodide in water.

11.15. Calculate the solubility (mol/L) of copper(I) iodide in a 0,50 mol/L aqueous solution of potassium iodide. Consider the following additional equilibrium that is established

$$Cu^{+}(ac) + 2I^{-}(ac) \rightleftharpoons [CuI_2^{-}](ac) \qquad K_f = 7{,}1 \cdot 10^{8}$$

11.16. Explain why copper(I) ions are unstable in aqueous solution using electrode potential values (quantitative) and the Theory of Hard and Soft Acids and Bases (qualitative).

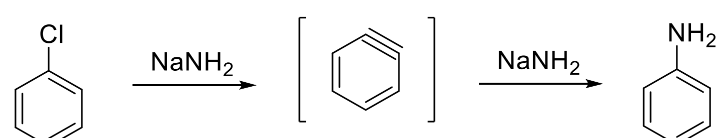
The dissociation constants of ascorbic acid are pKa1 $pK_{a1} = 4{,}1$ y $pK_{a1} = 11{,}8$ $pK_{ps}(CuI) = 11{,}9$

E°(V): $S_4O_6^{2-}/S_2O_3^{2-} = 0{,}08$; $I_2/I^- = 0{,}54$; $IO^-/I^- = 0{,}49$; $HIO/I_2 = 1{,}45$; $Cu^{2+}/Cu^+ = 0{,}15$; $Cu^{2+}/Cu = 0{,}34$

Problem 12. Impossible molecules (20% of total; only 12th grade)

12.1	12.2	12.3	12.4	12.5	12.6	12.7	12.8	12.9	12.10	12.11	12.12	12.13
4	4	6	12	2	2	2	6	2	6	2	3	3
12.14	12.15	12.16	12.17	12.18	12.19	12.20	12.21	12.22	12.23	12.24	Total	Total
2	2	2	3	2	6	2	2	3	2	---	80	20

Part I. Benzene derivatives can be converted into a reactive intermediate called benzyne. One way to form a benzyne is the treatment of chlorobenzene with $NaNH_2$. In this case, the benzyne forms aniline under the same conditions.



benzyne intermediate

12.1. Propose a mechanism for the formation of aniline from chlorobenzene.

Professor Rolando (also known as *Rolo* to his friends) is the most senior member of the National Pre-Selection and his passion is Organic Chemistry. A student once asked *Rolo* to obtain sodium amide to synthesize aniline. Under strictly anhydrous conditions, in an inert glove compartment, *Rolo* introduced a tiny bit of sodium into liquid nitrogen at −33°C. An intense blue color was immediately obtained, but only after adding a pinch of iron(III) nitrate was intense bubbling observed and the solution began to discolor.

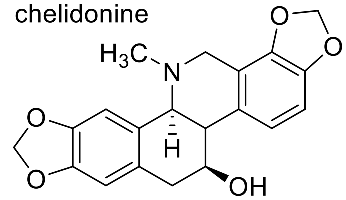
EXAM CODE (only for the use of the code manager)

12.2. Formulate chemical equations that explain the above results.

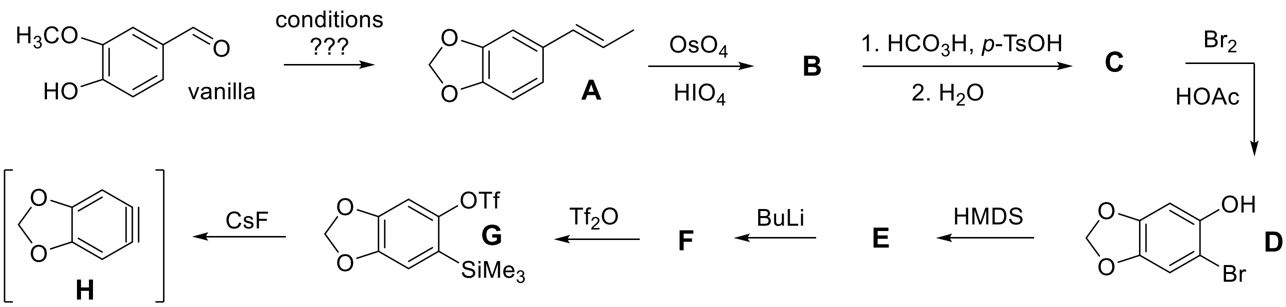
Then, a student asked *Rolo* if benzenes had any other applications besides obtaining aniline. It turns out that benzenes can also be formed into complex molecules, and *Rolo* discovered that a benzene intermediate is part of the key step for the total synthesis of the natural product chelidonine. The synthetic sequence in question is discussed below.

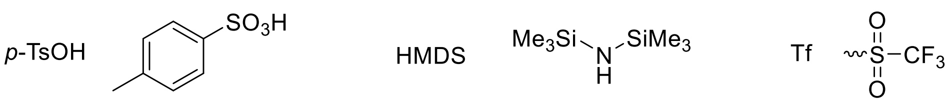
Starting material **A** is oxidized first with OsO_4 and HIO_4. This combination results in the same oxidation as neutral ozonolysis, giving **B**. **B** is further oxidized in hydrolyzing conditions to provide **C**. **C** is brominated under acidic conditions to give monobrominated compound **D**. **A** sequence of silylating reagent HMDS, butyllithium, and triflic anhydride (Tf_2O) result in benzyne precursor **G**. **G** is converted into the reactive benzyne **H** using CsF.

chelidonine



12.3. Propose a sequence that allows obtaining **A** from the natural product vanillin.





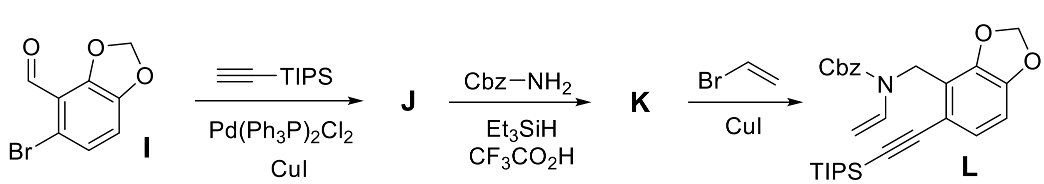
12.4. Draw the structures of **B**, **C**, **E** and **F**.

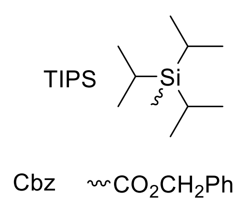
12.5. Give the name of the oxidation from **B** to **C**.

12.6. Explain the regioselectivity of monobromination to give **D**.

12.7. Propose a mechanism for the formation of benzyne **H** from **G**.

The benzyne reacts with a separate compound **L**. The synthesis of **L** proceeds as shown below. A Sonogashira coupling with **I** gives **J**. Treatment of J with $CbzNH_2$ in the presence of Et_3SiH and trifluoroacetic acid gives **K**. Another metal-catalyzed coupling gives the reaction partner **L**.



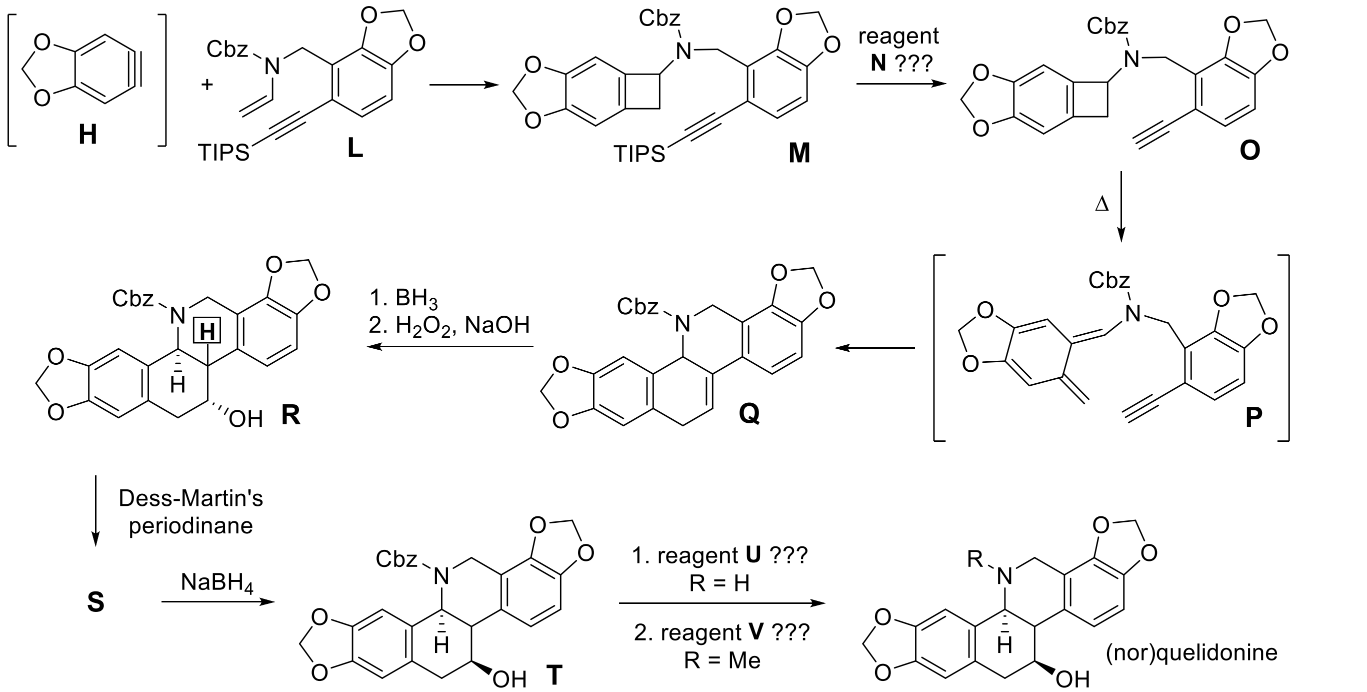


12.8. Draw the structures of **J** and **K**.

12.9. Mention the role of Et_3SiH in the conversion of **J** to **K**.

Next, *in situ* formed benzyne **H** and molecule **L** undergo a cascade to give a polycyclic system **M**. The final steps of the synthesis convert **Q** into norchelidonine. Borane followed by basic H_2O_2 gives alcohol **R**. Treatment with Dess-Martin periodinane converts **R** into **S**, which is followed by reaction with $NaBH_4$ to provide **T**. A single step converts **T** into the natural product norchelidonine and a second step in chelidonine.

EXAM CODE (only for the use of the code manager)



12.10. Propose the reagents **N**, **U** and **V**.

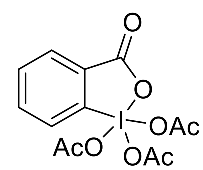
12.11. Give the name of the reaction that converts **P** into **Q**.

12.12. Draw **R** showing the stereochemistry of the hydrogen atom highlighted inside a square.

12.13. Draw the structure of **S**.

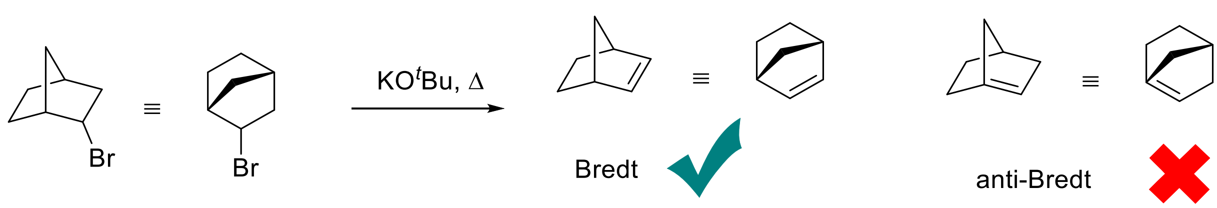
12.14. What is the function of the Dess-Martin periodinane?

12.15. What isomeric relationship exists between **R** and **T**?



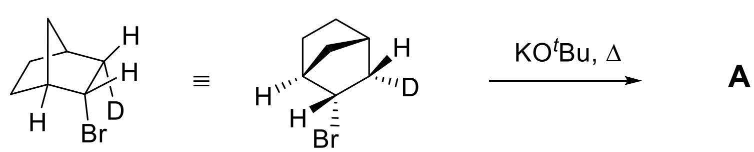
Dess-Martin's periodinane

Part II. When Rolo explains bridged bicycles to his students, he always teaches them Bredt's rule: "A double bond cannot be located in the bridgehead position unless the rings are large enough", typically with eight or more members. Elimination reactions in these compounds generally obey Bredt's rule, as illustrated below.



12.16. What is the cause of Bredt's rule?

12.17. Consider the elimination in the following isotopically labeled compound. Predict the product of the reaction, clearly indicating the isotopes present in the product.



To Rolo's surprise, in 2024, McDermott and coworkers developed a way to prepare the anti-Bredt alkene. Selenium is added to enone **B** to provide **C**. Treatment with excess Grignard reagent gives compound **D**, which can undergo olefin metathesis with Grubbs catalyst to provide cyclic compound **E**. **E** undergoes radical cyclization to give compound **F**, which is converted to **G** using Tf_2O in the presence of pyridine.

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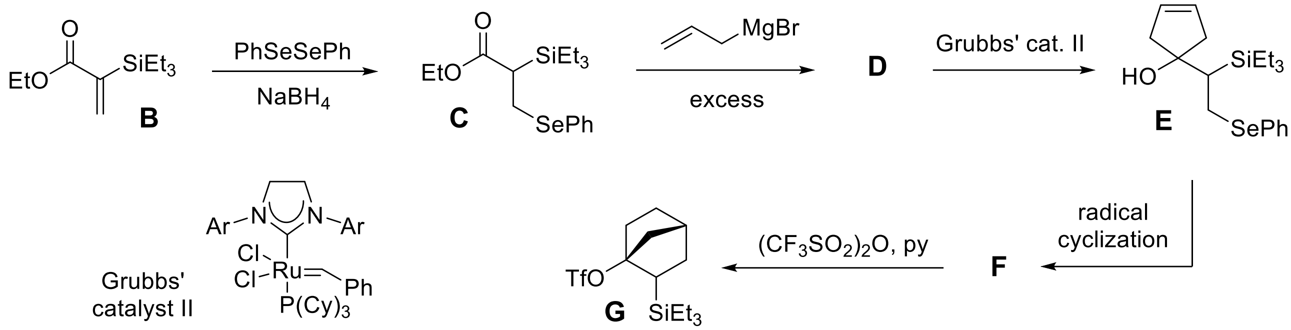
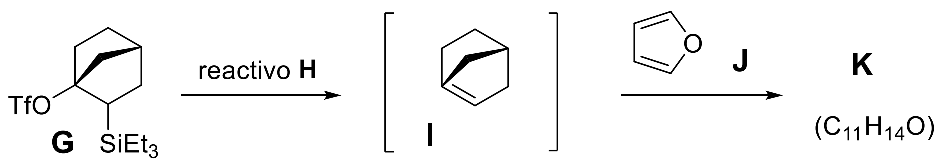


12.18. State whether selenides are hard or soft nucleophiles. Explain your answer and explain how it affects the reaction from **B** to **C**.

12.19. Draw the structures **D** and **F**.

12.20. What is the oxidation number of ruthenium in Grubbs' catalyst?

G can be converted into the anti-Bredt alkene **I**; however, **I** is extremely reactive so the researchers provided a reaction partner **J**, so that **I** could react immediately.



12.21. Suggest a reagent **H** that allows obtaining the anti-Bredt alkene.

12.22. Represent the structure of the final product **K**.

12.23. In fact, **G** exists as two diastereomers (**G1** and **G2**), and only one of them can be converted into the anti-Bredt alkene. Based on the structural information provided, predict which isomer is able to form the alkene. Justify your answer.

ANSWER FROM HERE ON

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